From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - I

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have only one correct answer. Each question carries **+2 marks** for correct answer. There is no negative marking.

Section-A (11 to 15) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Single Correct Choice Type

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Two balls are dropped from the same point after an interval of 1 second. Their separation 3 second after the release of the second ball. ($g = 10\ m/s^2$)
   (A) 25 m (B) 30 m
   (C) 35 m (D) 40 m

2. The upper half of an inclined plane of inclination θ is perfectly smooth while the lower half is rough. A block starting from rest at the top of the plane will come to rest at the bottom if the coefficient of friction between the block and the lower half of the plane is
   (A) $\mu = 2\tan\theta$ (B) $\mu = \tan\theta$
   (C) $\mu = \frac{2}{\tan\theta}$ (D) $\mu = \frac{1}{\tan\theta}$

3. A spring is compressed between two blocks of masses m_1 and m_2 placed on a horizontal frictionless surface. When the blocks are released. The blocks travel distance x_1 and x_2 respectively before coming to rest. The ratio x_1/x_2 is
   (A) m_1/m_2 (B) m_2/m_1
   (C) $\sqrt{\frac{m_1}{m_2}}$ (D) $\sqrt{\frac{m_2}{m_1}}$

4. If A is the areal velocity of planet of mass M, its angular momentum is
   (A) M (B) 2MA
   (C) A^2M (D) AM^2

5. A meteor of mass M breaks up into two parts. The mass of one part is m. For a given separation r the mutual gravitational force between the two parts will be maximum if
   (A) m = (M/2) (B) m = (M/3)
   (C) $m = \frac{M}{\sqrt{2}}$ (D) $m = \frac{M}{2\sqrt{2}}$

Space for Rough work

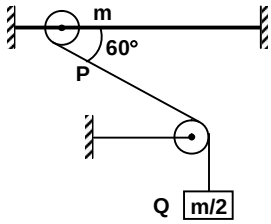
6. A wire breaks when subjected to a stress S. If ρ is the density of the material of the wire and g, the acceleration due to gravity, then the length of the wire so that it breaks by its own weight is
(A) ρgs (B) $\rho g/s$
(C) gs/ρ (D) $s/\rho g$

7. A cylinder has a radius r. To what height h should it be filled with water so that the thrust on its walls is equal to that on its bottom
(A) h = 2r (B) h = r
(C) h = (r/2) (D) h = (r/4)

8 A swimmer can swim in still water with a speed of $\sqrt{5}$ m/s. While crossing a river his average speed is 3 m/s. If he cross the river in the shortest possible time, what is the speed of flow of water?
(A) 2 m/s (B) 4 m/s
(C) 6 m/s (D) 8 m/s

9 A car starting from rest is accelerated at constant rate until it attains a constant speed v. It is then retarded at a constant rate until it comes to rest. Considering that the car moves with constant speed for half of the time of total journey, the average speed of the car for the journey is
(A) $\frac{v}{4}$ (B) $\frac{3v}{4}$
(C) $\frac{3v}{2}$ (D) Data insufficient

10. A smooth ring P of mass m can slide on a fixed horizontal rod. A string tied to the ring passes over a fixed pulley and carries a block Q of mass (m/2) as shown in the figure. At an instant, the string between the ring and the pulley makes an angle 60° with the rod.
The initial acceleration of the ring is



(A) $\frac{2g}{3}$ (B) $\frac{2g}{6}$
(C) $\frac{2g}{9}$ (D) $\frac{g}{3}$

Space for Rough work

Multiple Correct Answer(s) Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

11. A particle having mass m and positive charge q whirls along vertical circle of radius R. At the centre of circle, another positive charge 2q is fixed.
(A) The minimum velocity given at the lowest point to complete vertical circular motion is less than $\sqrt{5gR}$ for motion in gravity.
(B) For any value of velocity given in horizontal direction at the lowest point in gravity free space it will perform vertical circular motion.
(C) Tension in gravity free space cannot becomes zero at any point during motion.
(D) If the charge at the centre is removed and considering the motion in gravity tension will also be function of charge on particle.

12. The potential energy of a particle of mass 0.1 kg, moving along the x-axis, is given by $U = 5x(x-4)$J, where x is in meters. It can be concluded that
(A) The particle is acted upon by a variable force.
(B) The minimum potential energy during motion is -20 J
(C) The speed of the particle is maximum at x = 2m.
(D) The period of oscillation of the particle is $(\pi/5)$ sec.

Space for Rough work

13. A cubical block is floating in a liquid with one third of it's volume immersed in the liquid. When the whole system accelerates upwards with acceleration of g/2.
(A) The fraction of volume immersed in the liquid will change.
(B) The buoyancy force on the block will change.
(C) The buoyancy force will increase by 50 percent.
(D) The pressure in the liquid will increased.

14. Two concentric metallic shell's of radius R and 2R, out of which the inner shell is having charge Q and outer shell is uncharged. If they are connected with a conducting wire. Then,
(A) Q amount of charge will flow from inner to outer shell.
(B) Q/e number of electrons will flow from outer to inner shell, where e charge on electron.
(C) $\frac{KQ^2}{4R}$ amount of heat is produced in the wire
(D) $\frac{KQ^2}{2R}$ amount of heat is produced in the wire.

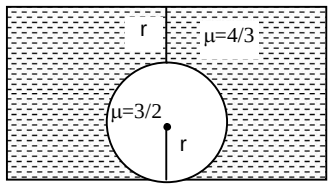
15. A sound wave of frequency n travels horizontally to the right. It is reflected from a large vertical plane surface moving to the left with a speed v. The speed of sound in the medium is c. Then,
(A) The number of waves striking the surface pre second is $\frac{n(c+v)}{c}$.
(B) The frequency of the reflected wave is $\frac{n(c+v)}{(c-v)}$
(C) The wavelength of the reflected wave is $\frac{c(c-v)}{n(c+v)}$
(D) The number of beats heard by a stationary listener to the left of the reflecting surface is $\frac{nv}{c-v}$

Space for Rough work

SECTION –C
Integer Answer Type

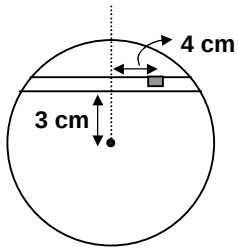
This section contains **5 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

1. A glass sphere having refractive index (3/2) is having a small irregularity at its centre. It is placed in a liquid of refractive index $\frac{4}{3}$ such that surface of liquid is r high above sphere where r is radius of sphere. If irregularity is viewed from above normally, the distance is 26/K cm from centre where eye will observe the irregularity. Then find the value of K. (r = 20 cm)



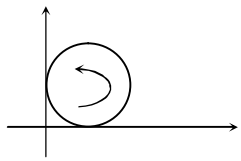
2. Two block of masses m and M are connected by means of a metal wire passing over a frictionless fixed pulley. The area of cross-section of the wire is 6.5×10^{-9} m^2 and its breaking stress is 2×10^9 Nm^{-2}. If m = 1 kg, if the maximum value of M is K × 1.86 kg for which the wire will not break. Find the value of K. ($g = 10\ m/s^2$)

3. A small cubical block of mass 1 kg is placed inside a rough rectangular groove $\left(\mu = \frac{1}{2}\right)$ made in a circular table as shown in the figure. The table starts rotating with angular acceleration 1 rad/sec^2 in a horizontal plane about its axis. The time is $10\sqrt{K}$ sec after which the blocks will start motion with respect to table. Find the value of K. (Take g = 10 m/s^2) Assume the size of block slightly smaller then the width of groove.



4. Two particles of mass M = 3 kg each are kept on a horizontal circular platform on two mutually perpendicular radii at equal distance R = 1 m from the centre of the table. The particles are connected with a string, which is just taught when the platform is not rotating. Coefficient of friction between the platform and block is μ = 0.1. Then the maximum angular speed (ω in rad/sec) of platform about its centre so that the blocks remain stationary relative to platform. (take g = 10 m/s^2)

5. A uniform cylinder of radius R(= 3m) is spin about its axis at an angular velocity ω_0 (= $40\sqrt{\pi}$ rad/sec) and placed between two perpendicular wall. The coefficient of friction between the walls and cylinder is μ(= 2). Then, 25 K turns will the cylinder make before it stops. Find the value of K.



Space for Rough work

Chemistry

PART – II

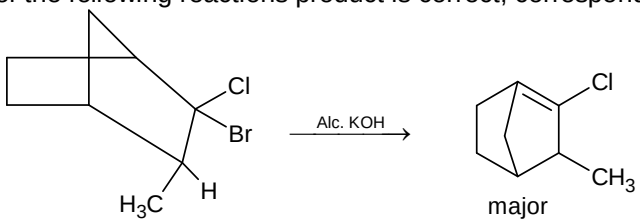
SECTION – A

Single Correct Choice Type

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

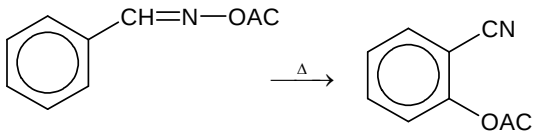
1. Which of the following reactions product is correct, correspondingly?

(A)

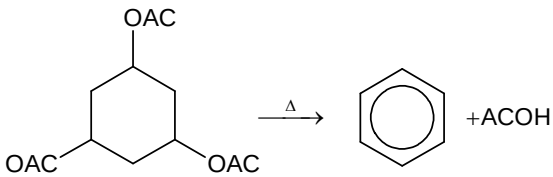


major

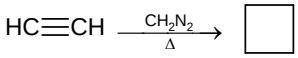
(B)



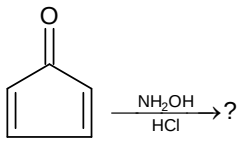
(C)



(D)

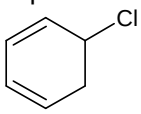


2.

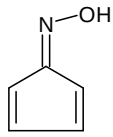


The major product of the following reaction is.....?

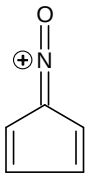
(A)



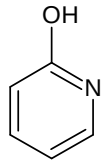
(B)



(C)

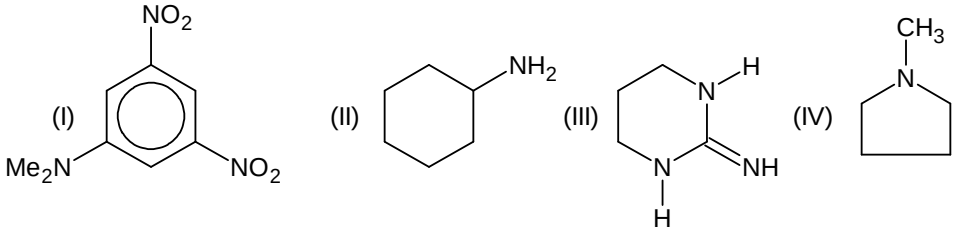


(D)



Space for Rough work

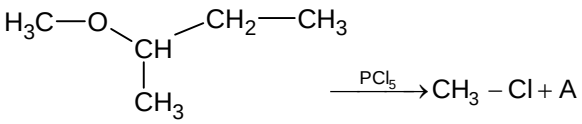
3. What is the correct order of basic strength of the following amines?

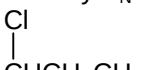


(A) (II) > (IV) > (III) > (I)
(B) (IV) > (I) > (III) > (II)
(C) (IV) > (I) > (II) > (III)
(D) (III) > (IV) > (II) > (I)

4. The complex $[Fe(H_2O)_5(NO)]^{2+}$ is formed in the ring test for nitrate when freshly prepared $FeSO_4$ solution is added to aqueous solution of NO_3^- followed by addition of conc. H_2SO_4. The complex is formed by charge transfer, which of the following change occurs?

(A) $Fe^{3+} \longrightarrow Fe^{2+}$; $NO \longrightarrow NO^+$
(B) $Fe^{2+} \longrightarrow Fe^{3+}$; $NO \longrightarrow NO^+$
(C) $Fe^{2+} \longrightarrow Fe^{+}$; $NO \longrightarrow NO^+$
(D) $Fe^{2+} \longrightarrow Fe^{3+}$; $NO^+ \longrightarrow NO$

5. 

(A) CH_3Cl is formed by S_N1 process.
(B) CH_3Cl is formed by S_N2 process
(C) A is  which is formed by S_N2 process
(D) A is  which is formed by S_N1 process

6. For the following equilibria in a close container

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

At a constant temperature, the volume of the container is halved suddenly. Which of the following is correct for K_p and α?

(A) Both K_p and α changes
(B) None of the K_p and α change
(C) K_p changes but α does not
(D) α - changes but K_p does not

Space for Rough work

7. Aluminum chloride exists as a dimer Al_2Cl_6 in solid state as well as in solution of non-polar solvents such as C_6H_6 . When dissolved in water, it gives

(A) $Al_2O_3 + 6HCl$ (B) $\left[Al\left(H_2O\right)_6\right]^{+} + 3Cl^{-}$

(C) $\left[\mathrm{Al(OH)_6}\right]^{3-} + 6\mathrm{HCl}$ (D) $Al^{3+} + 3Cl^{-}$

8. Resonance Energy of Toluene is 37.5 kcal/mol while that of Benzene is 36 kcal/mole then which of the following explains this difference
(A) Zeeman effect (B) Electromeric effect
(C) Hyper-conjugation (D) – I effect

9. The ortho and para-hydrogens possess:
(A) Same physical properties but different chemical properties.
(B) Different physical properties but same chemical properties.
(C) Same chemical and physical properties.
(D) Different physical and chemical properties.

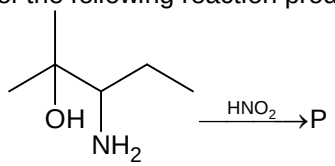
10. During extraction of Fe
(A) Carbon is ultimately converted to CO_2 in combustion zone in blast furnace.
(B) CO_2 is one of the by-products of reduction zone of blast furnace.
(C) Cast iron is obtained from blast furnace.
(D) Uppermost zone of blast furnace is slag formation zone.

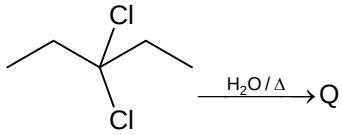
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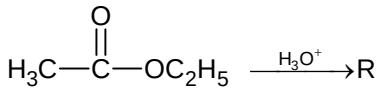
Multiple Correct Answer(s) Type

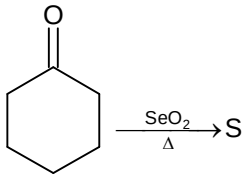
This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

11. Which of the following reaction products give haloform test?

(A) 

(B) 

(C) 

(D) 

12. Which of the following statement(s) is/are correct?
(A) Anti conformation of $H_2C(NH_2)—CH_2—NH_2$ is always more stable than its Gauche conformation.
(B) Gauche conformation of $H_2C(OH)—CH_2F$ is more stable than anti-conformation at room temperature.
(C) On increasing temperature dipole moment of pure $Cl—CH_2—CH_2—Cl$ increases.
(D) In case of 1, 4 – dihydroxycyclohexane twist boat conformer is most stable.

13. Which of the following statement(s) is/are incorrect?
(A) Manganese salts give a violet borax bead test in reducing flame.
(B) From mixed salt of AgCl and AgI, NH_4OH solution dissolves only AgCl.
(C) Fe^{3+} ions given a deep green precipitate, on adding $K_4[Fe(CN)_6]$ solution.
(D) On strong heating $Na_2S_2O_3.5H_2O$, a polysulphide is resulted.

14. Which of the following is / are correct for amino acids?
(A) At pI, there is no net charge on amino acid.
(B) At pH > pI, there is net negative charge on amino acid.
(C) Amino acids are amphoteric in nature.
(D) They are highly soluble in benzene like organic solvents.

15. Which of the following is / are correct?
(A) All Arrhenius acids are also Bronsted acids.
(B) All Bronsted bases are also Arrhenius bases.
(C) All Bronsted bases are also Lewis bases.
(D) pH of 10^{-8} M HCl (aq) is 7.04 while that of 10^{-8} M NaOH (aq) is 6.96.

Space for Rough work

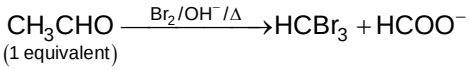
SECTION –C
Integer Answer Type

This section contains **5 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

1. The osmotic pressure of 0.1 M monobasic acid solution is 0.11 RT at 'T Kelvin', what is its pH value?

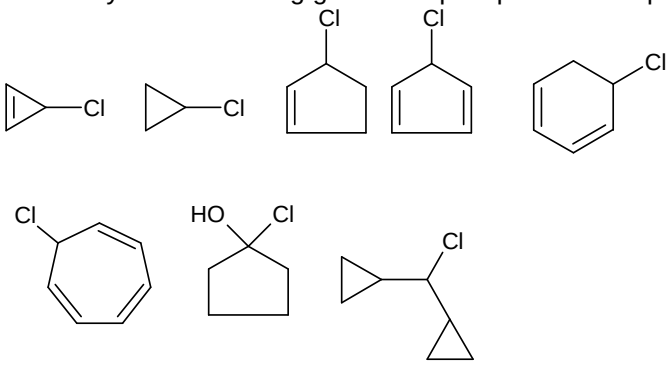
2. How many equivalents of HCHO are formed by oxidative cleavage of one equivalent of D-glucose by excess of HIO_4 in water?

3. How many equivalents of base are consumed in the following reaction?



4. How many of the following salts are white and soluble in dilute HNO_3?
$BaCrO_4$, Hg_2CrO_4, ZnS, $BaSO_4$, BaS_2O_3, CH_3COOAg, $AgNO_2$

5. How many of the following give white precipitate with aqueous $AgNO_3$ solution?



Space for Rough work

Mathematics

PART – III

SECTION – A

Single Correct Choice Type

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Number of solutions of the equation [y + [y]] = 2 cosx is, where y = $\frac{1}{3}$[sinx + [sinx + [sinx]]] and [.] denotes the greatest integer function
   (A) 1 (B) 2
   (C) 3 (D) none of these

2. The differential equation of the system of circles touching the x-axis at origin is
   (A) $(x^2 - y^2)\frac{dy}{dx} - 2xy = 0$ (B) $(x^2 - y^2)\frac{dy}{dx} + 2xy = 0$
   (C) $(x^2 + y^2)\frac{dy}{dx} - 2xy = 0$ (D) $(x^2 + y^2)\frac{dy}{dx} + 2xy = 0$

3. If α, β, γ, δ are the smallest positive angles in ascending order of magnitude which have their sines equal to the positive quantity k, then the value of $4\sin\frac{\alpha}{2} + 3\sin\frac{\beta}{2} + 2\sin\frac{\gamma}{2} + \sin\frac{\delta}{2}$ is equal to
   (A) $2\sqrt{1-k}$ (B) $2\sqrt{1+k}$
   (C) $2\sqrt{k}$ (D) none of these

4. If equation $x^3 - x + a^2 - a + \frac{2}{3\sqrt{3}} < 0$ is true for atleast one positive x, then a belongs to
   (A) (0, 1) (B) (1, ∞)
   (C) (2, $3\sqrt{3}$) (D) none of these

Space for rough work

5. Number of solutions of the equation $\sqrt{3x-2}+\sqrt{4x+4}=\sqrt{5x+1}+\sqrt{4x-5}$ is
(A) 0 (B) 1
(C) 2 (D) 4

6. The value of $\lim_{n\to\infty}\frac{1}{n^{n\lambda}}\left[(n+1)^{\lambda}(n+2)^{\lambda}...(n+n)^{\lambda}\right]^{1/n}$ is equal to
(A) $\left(\frac{4}{e}\right)^{\lambda}$ (B) $\left(\frac{1}{e}\right)^{\lambda}$
(C) $\left(\frac{3}{e}\right)^{\lambda}$ (D) $\left(\frac{2}{e}\right)^{\lambda}$

7. If $0 \le \arg z \le \frac{\pi}{4}$, then the least value of $\sqrt{2}\,|2z-4i|$ is
(A) 6 (B) 1
(C) 4 (D) 2

8. If α, β, γ be the roots of the equation $2x^3 + 3x^2 - 12x + 3 = 0$ and A(α, β, γ), B(β, γ, α), C(γ, α, β) represent vertices of a triangle ABC then the centroid of the triangle lies upon the line
(A) x = y = z (B) x = 2y = 3z
(C) x = – 2y = 3z (D) x = y = – 2z

9. If $a = \sqrt{\cot^{-1}x}$, $b = \cot^{-1}x$ and $c = (\cot^{-1}x)^2$ and a > b > c then x lies in the interval
(A) (0, 1) (B) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(C) (cot1, ∞) (D) (– cot1, 1)

10. Given a triangle ABC with circumcentre at origin. If $a \le 4 \sin A$ then for any point (a, y) inside the circumcircle of ΔABC
(A) |xy| < 2 (B) 2 < |xy| < 4
(C) $\frac{1}{2} < |xy| < \frac{3}{4}$ (D) |xy| > 4

Space for rough work

Multiple Correct Answer(s) Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

11. If in a ΔABC a, b and $\angle A$ are given and C_1, C_2 are the possible values of the third side, then
(A) $C_1 + C_2 = 2b \cos A$ (B) $C_1 \cdot C_2 (b^2 - a^2)$
(C) $C_1^2 + C_2^2 - 2C_1C_2 \cos 2A = 4a^2 \cos^2 A$ (D) $C_1 + C_2 = 2a \cos B$

12. $\frac{1}{\sqrt{2}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{8}} + \frac{1}{\sqrt{8}+\sqrt{11}} + \cdots$ n terms value of the above expression is
(A) $\frac{\sqrt{3n+2}-\sqrt{2}}{3}$ (B) $\frac{n}{\sqrt{2+3n}+\sqrt{2}}$
(C) less than n (D) less than $\sqrt{\frac{n}{3}}$

13. If inside a big circle exactly 24 small circles, each of radius 2, can be drawn in such a way that each small circle touches the big circle and also touches both its adjacent small circles, then radius of the big circle is
(A) $2\left(1+\operatorname{cosec}\frac{\pi}{24}\right)$ (B) $\left(\frac{1+\tan\frac{\pi}{24}}{\cos\frac{\pi}{24}}\right)$
(C) $2\left(1+\operatorname{cosec}\frac{\pi}{12}\right)$ (D) $\frac{2\left(\sin\frac{\pi}{48}+\cos\frac{\pi}{48}\right)^2}{\sin\frac{\pi}{24}}$

14. If $x^2 + x + 1 - |[x]| = 0$ (where [.] denotes greatest integer function) then
(A) x = 1 (B) x = – 1
(C) exactly two value of x in [–2, – 1] (D) none of these

15. If acosx + cotx + 1 = cosecx then the possible values of x can be
(A) $x = n\pi$, $a \in R$, $n \in I$
(B) $x = (4n+1)\frac{\pi}{2}$, $a \in R$, $n \in I$
(C) $x = \frac{1}{2}\sin^{-1}\left(\frac{4(a+1)}{a^2}\right)$, $a \in (-\infty, 2-2\sqrt{2}\,] \cup [2+2\sqrt{2}\,, \infty) - \{-1\}$
(D) none of these

Space for rough work

SECTION –C
Integer Answer Type

This section contains **5 questions.** The answer to each question is a **single digit integer**, ranging from 0 to 9 (both inclusive).

1. If $a = \sum_{r=1}^{\infty} \frac{1}{r^2}$, $b = \sum_{r=1}^{\infty} \frac{1}{(2r-1)^2}$, then $\frac{3a}{b}$ is equal to _____

2. Let f : R $\to$ R defined by f(x) = sin x + x if $\int_0^{\pi} f^{-1}(x)dx = \frac{\pi^2}{2} - k$, then the value of k is _____

3. The figures 4, 5, 6, 7, 8 are written in every possible order. The number of numbers greater than 56000 is 15K then K is equal to _____

4. Let y = g(x) be the image of f(x) = x + sin x about the line x + y = 0. If the area bounded by y = g(x), x-axis, x = 0 and x = 2π is A, then $\frac{A}{\pi^2}$ is _____

5. If $|\vec{a}| = 2$, $|\vec{b}| = 3$, and $|\vec{c}| = 4$ then $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 \le (12K + \lambda)$ then λ is equal to _____

Space for rough work